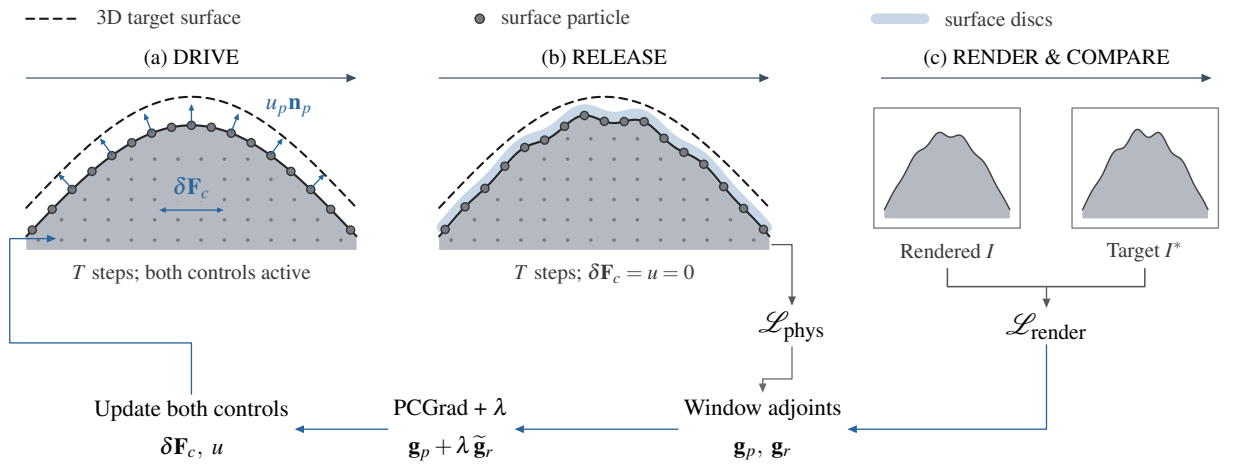

3D target surface
surface particle
surface discs
(a) DRIVE
(b) RELEASE
(c) RENDER & COMPARE
$u_p \mathbf{n}_p$
$\delta \mathbf{F}_c$
T steps; both controls active
T steps; $\delta \mathbf{F}_c = u = 0$
Rendered I
Target I^*
$\mathscr{L}_{\text{phys}}$
$\mathscr{L}_{\text{render}}$
Update both controls
$\delta \mathbf{F}_c,\ u$
PCGrad + λ
$\mathbf{g}_p + \lambda\, \tilde{\mathbf{g}}_r$
Window adjoints
$\mathbf{g}_p,\ \mathbf{g}_r$
Differentiate through driven and released steps